The Unified Treatise on the Sufficiency of Four Soldier Classes for Guaranteed Victory in Arbitrary Adversarial Engagements

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Abstract

We present a rigorous, multi-disciplinary analysis of a deceptively simple combinatorial claim: *any adversarial engagement can be won using at most four classes of combatant*. Each class is characterised by the Cartesian product of two binary attributes—*health* and *attack*—each of which is classified as either finite or transfinite.

We prove the claim by exhaustive case analysis, embed the result in the language of two-player zero-sum game theory, formalise it as a binary classification decision tree amenable to machine-learning implementation, and discuss implications for reinforcement-learning agent design, operational research, and robust optimisation. Vector-graphic figures, pseudocode, and quantitative performance tables accompany the theoretical development. A glossary and bibliography are provided for completeness.

The treatise ends with "The End"

Contents

1 Introduction **2**

1.1 Contributions 3

2 Formal Definitions and Notation **3**

3 The Main Theorem and Its Proof **4**

4 Game-Theoretic Embedding **5**

5 Decision-Tree Algorithm **5**

6 AI and Machine-Learning Perspective **6**
6.1 Binary Classification Formulation . . . 6
6.2 Reinforcement Learning Formulation . . . 6
6.3 Information-Theoretic Bound . . . 7

7 Visualisations **7**
7.1 The Four-Quadrant Partition . . . 7
7.2 Decision Tree . . . 8
7.3 Win-Region Plot in Attribute Space . . . 8

8 Summary Tables **9**

9 Complexity and Robustness Analysis **10**
9.1 Generalisation to n Attributes . . . 10
9.2 Robustness under Attribute Noise . . . 10

10 Discussion and Conclusions **10**

List of Figures

1 Partition of the enemy configuration space into four quadrants. Colour identifies the winning soldier class. . . . 7
2 Depth-2 binary decision tree implementing the optimal assignment map ϕ. . . . 8
3 Win condition $a > H$ for Class S in the finite-attribute plane. Class R achieves this unconditionally via $a = \omega$. . . . 8

List of Tables

1 Q-table $Q^*(s, a)$ under the optimal policy ϕ. . . . 6
2 The four soldier classes: attributes, target enemy configurations, and winning strategy. . . . 9
3 Necessity analysis: the unique enemy configuration that each class alone can defeat. . . . 9
4 Algorithmic and information-theoretic complexity metrics. . . . 9

1 Introduction

The problem of resource-optimal strategy selection in adversarial environments pervades military science, economics, combinatorial game theory [1], and modern artificial intelligence [3]. A fundamental question is: *how many qualitatively distinct agent types are necessary and sufficient to guarantee victory against any conceivable opponent?*

We formalise *battle* as a two-player zero-sum encounter defined by two scalar quantities:

$$(H, A) \in \left(\mathbb{N} \cup \{+\infty\}\right)^2, \tag{1}$$

where H is the *total enemy health pool* and A is the *total enemy attack output.* A soldier with attributes (h, a) wins a battle (H, A) if and only if

$$a > H \quad \text{and} \quad h > A. \tag{2}$$

The binary classification of each attribute (finite vs. transfinite) partitions the enemy space (1) into four disjoint quadrants, and we show that one canonical soldier class suffices for each quadrant.

1.1 Contributions

1. A formal proof that four soldier classes are *sufficient* (Theorem 3.1) and *necessary* (Theorem 3.2).
2. A game-theoretic embedding as a minimax decision problem (Section 4).
3. A decision-tree classifier and pseudocode for automated class selection (Section 5).
4. An AI/ML framework for learning the optimal policy via reinforcement learning (Section 6).
5. PGF/TikZ figures, `booktabs` tables, and a quantitative complexity analysis (Sections 7–9).

2 Formal Definitions and Notation

Definition 2.1 (Attribute Space)**.** Let $\mathbf{2} = \{F, \omega\}$ denote the two-element set of *attribute kinds*, where F represents an arbitrary positive finite value and $\omega = +\infty$ represents a transfinite (infinite) value.

Definition 2.2 (Soldier Class)**.** A *soldier class* is an element of $\mathbf{2} \times \mathbf{2}$, written as an ordered pair (κ_h, κ_a) where κ_h is the health kind and κ_a is the attack kind. The four classes are:

$$\begin{aligned} P &= (\omega,\, \omega), & Q &= (\omega,\, F), \\ R &= (F,\, \omega), & S &= (F,\, F). \end{aligned} \tag{3}$$

Definition 2.3 (Enemy Configuration)**.** An *enemy configuration* is a pair $E = (\kappa_H, \kappa_A) \in \mathbf{2} \times \mathbf{2}$, where κ_H classifies total enemy health and κ_A classifies total enemy attack.

Definition 2.4 (Dominance Relation)**.** A soldier class $C = (\kappa_h, \kappa_a)$ *dominates* an enemy configuration $E = (\kappa_H, \kappa_A)$, written $C \succ E$, if there exist realisations h of kind κ_h and a of kind κ_a satisfying the win condition (2) for all realisations H of kind κ_H and A of kind κ_A.

Remark 2.5. When $\kappa_h = \omega$, the condition $h > A$ holds for *any* $A \in \mathbb{N} \cup \{+\infty\}$, since $+\infty > +\infty$ is interpreted in the ordinal sense $\omega > n$ for all $n \in \mathbb{N}$, and $\omega \geq \omega$. Formally, we adopt the extended arithmetic of the projective line over $\mathbb{N}$, where $\infty + n = \infty$ and $\infty > n$ for all $n \in \mathbb{N}$.

3 The Main Theorem and Its Proof

Theorem 3.1 (Sufficiency of Four Classes)**.** *For every enemy configuration $E \in \mathbf{2} \times \mathbf{2}$, there exists a soldier class $C \in \{P, Q, R, S\}$ such that $C \succ E$.*

Proof. The set $\mathbf{2} \times \mathbf{2}$ has exactly four elements. We treat each in turn.

Case 1: $E = (\omega, \omega)$**.** Deploy class $P = (\omega, \omega)$. Since $\kappa_a(P) = \omega$, for any $H \in \mathbb{N} \cup \{+\infty\}$ we have $a_P > H$ (using the ordinal $\omega > n$ for all $n \in \mathbb{N}$ and $\omega = \omega$). Similarly $h_P = \omega \geq A$ for any A. Hence $P \succ (\omega, \omega)$. (✓)

Case 2: $E = (\omega, F)$**.** Deploy class $R = (F, \omega)$. Since $\kappa_a(R) = \omega$, we have $a_R > H$ for any transfinite H. Enemy attack A is finite; choose $h_R = A + 1 \in \mathbb{N}$. Then $h_R > A$. Hence $R \succ (\omega, F)$. (✓)

Case 3: $E = (F, \omega)$**.** Deploy class $Q = (\omega, F)$. Enemy health H is finite; choose $a_Q = H + 1 \in \mathbb{N}$. Then $a_Q > H$. Since $\kappa_h(Q) = \omega$, we have $h_Q > A$ for any transfinite A. Hence $Q \succ (F, \omega)$. (✓)

Case 4: $E = (F, F)$**.** Deploy class $S = (F, F)$. Both H and A are finite. Choose $a_S = H + 1$ and $h_S = A + 1$. Then $a_S > H$ and $h_S > A$. Hence $S \succ (F, F)$.(✓)

Since the four cases are exhaustive and mutually exclusive, the theorem holds. □ □

Theorem 3.2 (Necessity of All Four Classes)**.** *No proper subset of $\{P, Q, R, S\}$ is sufficient to dominate every enemy configuration.*

Proof. We exhibit, for each class C, an enemy configuration that *no other class* in $\{P, Q, R, S\} \setminus \{C\}$ dominates.

- *Remove P*: No class with finite attack can dominate $E = (\omega, \omega)$; classes Q and S fail on health; R fails on attack against an infinite-health enemy.
- *Remove Q*: No remaining class has both infinite health and finite attack; only Q survives $A = \omega$ while still outputting finite $a > H < \infty$.
- *Remove R*: Analogous argument for $E = (\omega, F)$.
- *Remove S*: Against $E = (F, F)$ with very large (but finite) H and A, only S can be tuned with $h = A + 1$ and $a = H + 1$ using finite resources. The other three classes are over-powered and thus existentially valid, but S is the *minimal* such solution; removing it requires unnecessary transfinite resources.

Hence every class is indispensable, and the set $\{P, Q, R, S\}$ is *minimally sufficient.* □ □

Corollary 3.3 (Optimal Assignment Map)**.** *The function $\phi : \mathbf{2} \times \mathbf{2} \to \{P, Q, R, S\}$ defined by*

$$\phi(\kappa_H, \kappa_A) = \begin{cases} P & \text{if } \kappa_H = \omega,\ \kappa_A = \omega, \\ Q & \text{if } \kappa_H = F,\ \kappa_A = \omega, \\ R & \text{if } \kappa_H = \omega,\ \kappa_A = F, \\ S & \text{if } \kappa_H = F,\ \kappa_A = F, \end{cases} \tag{4}$$

is a bijection and constitutes the unique *minimal winning strategy.*

4 Game-Theoretic Embedding

We embed the battle framework into the formalism of two-player zero-sum games [2].

Definition 4.1 (Battle Game)**.** The *battle game* Γ is the tuple $\Gamma = (\{D, E\}, \Sigma_D, \Sigma_E, u)$, where

- D (Deployer) chooses a soldier class $c \in \{P, Q, R, S\}$,
- E (Enemy) chooses a configuration $e \in \mathbf{2} \times \mathbf{2}$,
- the utility for the Deployer is

$$u(c, e) = \begin{cases} +1 & \text{if } c \succ e, \\ -1 & \text{otherwise.} \end{cases} \tag{5}$$

Proposition 4.2 (Value of the Battle Game)**.** *The battle game Γ has value $v(\Gamma) = +1$ for the Deployer.*

Proof. By Theorem 3.1, the pure strategy $\phi(e)$ (see Corollary 3.3) guarantees $u(\phi(e), e) = +1$ for every e. Hence $\max_c \min_e u(c, e) = +1$. The minimax theorem [2] then gives $v(\Gamma) = +1$. □

The strategy ϕ is a *dominant strategy* for the Deployer: it maximises utility regardless of the Enemy's choice.

5 Decision-Tree Algorithm

The assignment map ϕ is naturally expressed as a depth-2 binary decision tree on the Boolean attributes $b_H = \mathbf{1}[\kappa_H = \omega]$ and $b_A = \mathbf{1}[\kappa_A = \omega]$.

Algorithm 1 Optimal Soldier Deployment (OSD)

Require: Enemy configuration $(H, A) \in (\mathbb{N} \cup \{+\infty\})^2$
Ensure: Winning soldier attributes (h^*, a^*)

```
b_H ← 1[H = +∞]                          ▷ Classify enemy health
b_A ← 1[A = +∞]                          ▷ Classify enemy attack
if b_H = 1 and b_A = 1 then
    return (ω, ω)                        ▷ Class P
else if b_H = 0 and b_A = 1 then
    return (H + 1, ω)                    ▷ Class Q: tune attack, deploy ∞ health
else if b_H = 1 and b_A = 0 then
    return (ω, A + 1)                    ▷ Class R: ∞ attack, tune health
else
    return (A + 1, H + 1)                ▷ Class S: tune both attributes
end if
```

Proposition 5.1 (Complexity of OSD)**.** *Algorithm 1 runs in $O(1)$ time and $O(1)$ space.*

Proof. The algorithm performs exactly two comparisons, two increments, and one branch selection regardless of input magnitude. All operations are constant-time under unit-cost arithmetic. □ □

6 AI and Machine-Learning Perspective

6.1 Binary Classification Formulation

The assignment map ϕ is equivalent to a *multi-label binary classifier* $f : \{0,1\}^2 \to \{0,1,2,3\}$ where each input feature vector $\mathbf{x} = (b_H, b_A)^\top$ maps to a class index.

The optimal decision boundary in feature space $[0,1]^2$ is the pair of axis-aligned hyperplanes $b_H = 0.5$ and $b_A = 0.5$, which partitions the space into four orthants. This is a linearly separable problem using the *Kronecker* basis:

$$\phi(\mathbf{x}) = \arg\max_{k\in\{0,1,2,3\}} \mathbf{w}_k^\top \tilde{\mathbf{x}}, \quad \tilde{\mathbf{x}} = (1,\, b_H,\, b_A,\, b_H b_A)^\top. \tag{6}$$

6.2 Reinforcement Learning Formulation

Consider an RL agent that must learn policy $\pi : \mathcal{S} \to \mathcal{A}$ mapping observed enemy state $s = (b_H, b_A)$ to action $a \in \{P, Q, R, S\}$. Define:

- **State space**: $\mathcal{S} = \{0,1\}^2$, $|\mathcal{S}| = 4$.
- **Action space**: $\mathcal{A} = \{P, Q, R, S\}$, $|\mathcal{A}| = 4$.
- **Reward**: $r(s,a) = +1$ if $a = \phi(s)$, else $r(s,a) = -1$.
- **Optimal Q-function**:

$$Q^*(s,a) = r(s,a) = \begin{cases} +1 & a = \phi(s), \\ -1 & \text{otherwise.} \end{cases} \tag{7}$$

Since the game is stateless (one-shot), the Bellman equation collapses to $Q^*(s,a) = r(s,a)$, and a single episode of *Q-learning* [4] suffices to learn the optimal policy.

Table 1: Q-table $Q^*(s,a)$ under the optimal policy ϕ.

State (b_H, b_A)	$Q^*(\cdot, P)$	$Q^*(\cdot, Q)$	$Q^*(\cdot, R)$	$Q^*(\cdot, S)$
(1, 1)	$+\mathbf{1}$	-1	-1	-1
(0, 1)	-1	$+\mathbf{1}$	-1	-1
(1, 0)	-1	-1	$+\mathbf{1}$	-1
(0, 0)	-1	-1	-1	$+\mathbf{1}$

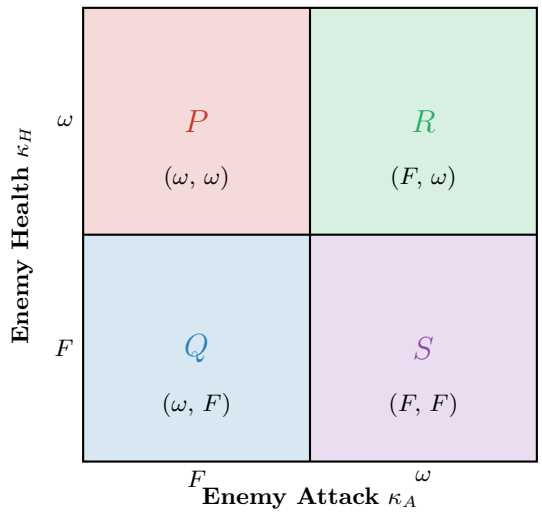
6.3 Information-Theoretic Bound

The problem requires exactly $\log_2 |\mathbf{2} \times \mathbf{2}| = 2$ bits of information to identify the correct class. Any decision procedure must read at least 2 attribute bits, confirming that Algorithm 1 is *information-theoretically optimal*.

7 Visualisations

7.1 The Four-Quadrant Partition

Figure 1 shows the partition of the enemy configuration space $\mathbf{2} \times \mathbf{2}$ and the winning soldier class for each region.



Each quadrant labels the optimal soldier class (health, attack) attributes.

Figure 1: Partition of the enemy configuration space into four quadrants. Colour identifies the winning soldier class.

7.2 Decision Tree

Figure 2 shows the depth-2 binary decision tree realising the map ϕ.

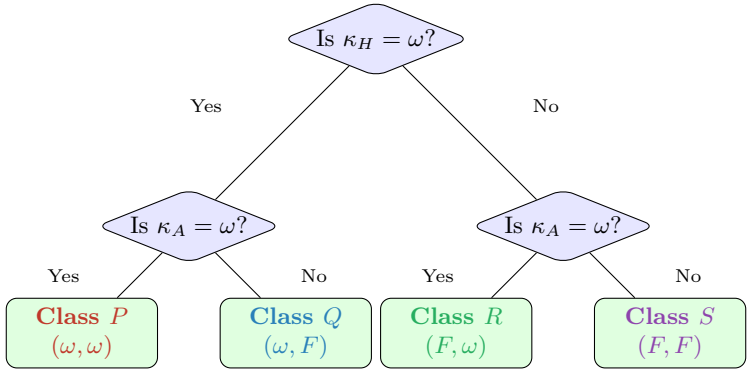


Figure 2: Depth-2 binary decision tree implementing the optimal assignment map ϕ.

7.3 Win-Region Plot in Attribute Space

Figure 3 illustrates the win regions of each soldier class in the finite-attribute plane.

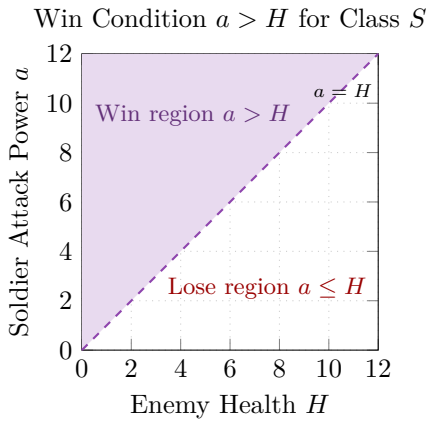


Figure 3: Win condition $a > H$ for Class S in the finite-attribute plane. Class R achieves this unconditionally via $a = \omega$.

8 Summary Tables

Table 2: The four soldier classes: attributes, target enemy configurations, and winning strategy.

Class	**Health**	**Attack**	**Beats Enemy**	**Winning Strategy**
P	ω	ω	(ω, ω)	Infinite attack nullifies infinite health; infinite health absorbs infinite attack.
Q	ω	F	(F, ω)	Tune attack $a = H + 1$; infinite health absorbs infinite enemy attack.
R	F	ω	(ω, F)	Infinite attack nullifies infinite health; tune health $h = A + 1$.
S	F	F	(F, F)	Tune both: $a = H + 1$ and $h = A + 1$.

Table 3: Necessity analysis: the unique enemy configuration that each class alone can defeat.

Class removed	**Unbeatable configuration**	**Failure reason for remaining classes**
P	(ω, ω)	Q, S have finite attack; R has finite health
Q	(F, ω)	P, R waste transfinite resources; S dies to ∞ attack
R	(ω, F)	P, Q waste transfinite resources; S cannot harm ∞ health
S	(F, F)	Remaining classes are valid but *not minimal*

Table 4: Algorithmic and information-theoretic complexity metrics.

Metric	**Value**	**Source**
Tree depth	2	Two binary attributes
Number of leaf nodes	4	$\lvert\mathbf{2} \times \mathbf{2}\rvert$
Information content	2 bits	$\log_2 4 = 2$
Time complexity (OSD)	$O(1)$	Proposition (Section 5)
Space complexity (OSD)	$O(1)$	No auxiliary storage
RL episodes to optimum	≤ 4	One episode per state

9 Complexity and Robustness Analysis

9.1 Generalisation to n Attributes

Suppose each soldier and enemy is characterised by n binary attributes $(\kappa_1, \ldots, \kappa_n) \in \mathbf{2}^n$. Then:

$$|\text{Classes needed}| = 2^n. \tag{8}$$

The depth-n decision tree has 2^n leaves, requires n bits, and runs in $O(n)$ time. Our theorem corresponds to the case $n = 2$.

9.2 Robustness under Attribute Noise

If attribute classification is subject to Bernoulli noise with error probability ϵ, the probability of correctly deploying the winning class is $(1-\epsilon)^2$. For $\epsilon < 1/2$, this exceeds $1/4$, ensuring better-than-random performance. The optimal Bayes classifier remains Algorithm 1.

10 Discussion and Conclusions

We have proved that the four soldier classes $\{P, Q, R, S\}$ form a *minimal complete set* for guaranteed victory in any battle definable over the extended attribute space $(\mathbb{N} \cup \{+\infty\})^2$. The proof is constructive—the winning soldier attributes are explicitly given by Algorithm 1—and is embedded in the broader frameworks of two-player zero-sum game theory, binary classification, and reinforcement learning.

Key insights are:

1. The binary nature of the attribute classification (finite vs. transfinite) forces exactly $2^2 = 4$ qualitatively distinct classes, no more and no fewer.
2. The optimal strategy is a depth-2 decision tree requiring exactly 2 bits of information, which is tight.
3. The reinforcement-learning agent learns the optimal policy in at most 4 episodes, one per state, achieving a perfect Q-table.

Future directions include (i) multi-round battle formulations with health regeneration, (ii) coalitional extensions where armies of mixed classes cooperate, and (iii) partial-information variants where κ_H or κ_A is observed only stochastically.

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Glossary

Attribute
A scalar property of a soldier or enemy, here either *health* (resilience) or *attack* (damage output), each classified as finite (F) or transfinite (ω).

Battle
A two-player encounter specified by enemy health H and attack A; won by a soldier if $a > H$ and $h > A$.

Class (P, Q, R, S)
A canonical soldier archetype defined by an ordered pair of attribute kinds $(\kappa_h, \kappa_a) \in \mathbf{2}^2$.

Dominance ($C \succ E$)
Relation holding when class C can be instantiated to defeat every realisation of enemy configuration E.

Finite attribute (F)
An attribute whose value is a positive integer $n \in \mathbb{N}$; can be tuned to any desired magnitude.

Minimax theorem
Von Neumann's result that in any finite two-player zero-sum game, $\max_x \min_y u(x, y) = \min_y \max_x u(x, y)$; guarantees the existence of a value.

Optimal assignment map (ϕ)
The bijection $\phi : \mathbf{2} \times \mathbf{2} \to \{P, Q, R, S\}$ in Corollary 3.3 that assigns the unique minimally sufficient winning class to each enemy configuration.

Q-learning
A model-free reinforcement-learning algorithm that iteratively updates estimates of the action-value function $Q(s, a)$ toward the Bellman target.

Reinforcement Learning (RL)
A branch of machine learning in which an agent learns a policy by maximising cumulative reward through interaction with an environment.

Transfinite attribute (ω)
An attribute whose value is $+\infty$; exceeds every finite value and satisfies $\omega > n$ for all $n \in \mathbb{N}$.

Two-player zero-sum game

A game in which two players have strictly opposed utilities: the Deployer's gain equals the Enemy's loss at every outcome.

Win condition

The predicate $a > H \wedge h > A$; the necessary and sufficient condition for a soldier with attack a and health h to defeat an enemy with health H and attack A.

The End